| QUALIFICATION REQUIREMENTS REMOVED |

MIL-M-38510/484A(USAF)
29 June 1984
SUPERSEDING
MIL-M-38510/484(USAF)
30 August 1982

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, N-CHANNEL, SILICON GATE, MONOLITHIC, PARALLEL INPUT/OUTPUT CONTROLLER

[INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION]

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic N-channel, silicon gate, parallel input/output controller. One product assurance class and a choice of lead finishes are provided, and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Frequency	Circuit
01	4.0 MHz	Parallel input/output controller
02	2.5 MHz	Parallel input/output controller

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, (see MIL-M-38510, appendix C)
Q	D-5 (40-lead, 9/16" x 2"), dual-in-line package

1.3 Absolute maximum ratings.

V_{CC} supply voltage range (referenced to ground)	-0.3 to +7.0 V dc
Voltage on any pin (referenced to ground)	-0.3 to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation:	
At -55°C	0.7 W
At 125°C	0.5 W
Lead temperature (soldering 10 seconds)	270°C
Maximum junction temperature (T_J)	
at T_C = +125°C	180°C
Thermal resistance, junction to case (θ_{JC})	40°C/W

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH}):	
Logic inputs	2.0 V dc
Clock input	V_{CC} - 0.6 V dc
Maximum low level input voltage (V_{IL}):	
Logic inputs	0.8 V dc
Clock input	0.45 V dc
Frequency of operation:	
Circuit 01	0.5 to 4.0 MHz
Circuit 02	0.5 to 2.5 MHz
Case operating temperature range (T_C)	-55° to +125°C
Clock rise time:	
Circuit 01	20 ns maximum
Circuit 02	30 ns maximum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specification and standard, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILTARY

MIL-M-38510 - Microcircuits, General Specification For.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510 and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Functional block diagram. The functional block diagram shall be as specified on figure 2.

3.2.3 Functional description, terms, and symbols. The functional description, terms, and symbols shall be as specified in 6.5.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510.

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TABLE I. Electrical performance characteristics.

Parameter	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Clock high input voltage	V_{IH1}		01,02	V_{CC} - 0.6	5.5	V
Clock low input voltage	V_{IL1}		"	-0.3	0.45	V
Logic input high voltage	V_{IH2}		"	2.0	---	V
Logic input low voltage	V_{IL2}		"	---	0.8	V
Logic low output voltage	V_{OL}	I_{OL} = 2.0 mA	"	---	0.4	V
Logic high output voltage	V_{OH}	I_{OH} = -250 µA	"	2.4	---	V
Power supply current	I_{CC}	Outputs open T_C = -55°C V_{CC} = 5.5 V	"	---	127	mA
Output leakage current low, open drain outputs	I_{LOL}	V_{IN} = 0.4 V	"	-10	+10	µA
Output leakage current high, open drain outputs	I_{LOH}	V_{IN} = 2.4 V	"	-10	+10	µA
Input low current (input and bidirectional)	I_{IL}	V_{IN} = 0.4 V	"	-10	+10	µA
Input high current (input and bidirectional)	I_{IH}	V_{IN} = 2.4 V	"	-10	+10	µA
Darlington drive current	I_{OH2}	V_{CC} = 4.5 V V_O = 1.5 V	"	-1.5	-3.8	mA
Maximum frequency 01 02	f_{MAX}		"	4.0 2.5	--- ---	MHz MHz
Clock input capacitance	C_{I1}		"	---	20	pF
Logic input capacitance	C_{I2}		"	---	5	pF
Output and bidirectional capacitance	C_O		"	---	10	pF

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le 125°C$	Figure no.	Reference no.	Device types 01 Min	01 Max	02 Min	02 Max	Unit
Clock cycle time	t_{cyc} 1/		6	1	250		400	---	ns
Clock pulse width High	t_{PWH1}		"	2	105	2,000	170	2,000	ns
Low	t_{PWL1}		"	3	105	2,000	170	2,000	ns
Clock time Fall	t_{fc}		"	4	---	20	---	30	ns
Rise	t_{rc}		"	5	---	20	---	30	
$\overline{CE}$, B/$\overline{A}$, C/$\overline{D}$ to $\overline{RD}$ $\overline{IORQ}$ ↓ setup time	t_{SHL1}		"	6	50 2/	---	50 2/	---	ns
Any hold times for specified setup time	t_{HLH1} t_{HHL1}		"	7	0	---	0	---	ns
$\overline{RD}$, $\overline{IORQ}$ to clock ↑ setup time	t_{SLH2} t_{SHL2}		"	8	115	---	115	---	ns
$\overline{RD}$, $\overline{IORQ}$ ↓ to data out delay	t_{PZL1} t_{PZH1}		"	9		380		430	ns
$\overline{RD}$, $\overline{IORQ}$ ↑ to data out float delay	t_{PLZ1} t_{PHZ1}		"	10		110	---	160	ns
Data in to clock ↑ setup time	t_{SZH1} t_{SZL1}		"	11	50	---	50	---	ns
$\overline{IORQ}$ ↓ to data out delay (INTACK cycle)	t_{PZL2} t_{PZH2}		"	12	250	---	340	---	ns
$\overline{MI}$ ↓ to clock ↑ setup time 3/	t_{SHL3}		"	13	90		210		ns
$\overline{MI}$ ↓ to clock ↑ setup time 3/ ($\overline{MI}$ cycle)	t_{SLH4}		"	14	0		0		ns
$\overline{MI}$ ↓ to IEO ↓ delay (interrupt-immediately preceeding $\overline{MI}$ ↓) 3/	t_{PHL1}		"	15		190		300	ns
IE1 to $\overline{IORQ}$ ↓ setup time (INTACK cycle)	t_{SHL5}		"	16	140		140		ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le 125°C$	Figure no.	Reference no.	Device types				Unit
					01		02		
					Min	Max	Min	Max	
IEI ↓ to IEO ↓ delay	t_{PHL2}		6	17		130		190	ns
IEI ↑ to IEO ↑ delay (after ED decode)	t_{PLH3}		"	18		160		210	ns
$\overline{IORQ}$ ↑ to clock ↓ setup time (to activate READY on next clock cycle)	t_{SLH6}		"	19	200		220		ns
Clock ↓ to READY ↑ delay	t_{PHL4}		"	20	190		200		ns
Clock ↓ to READY ↓ delay	t_{PHL5}		"	21	140		150		ns
$\overline{STROBE}$ pulse width	t_{PWL2}		"	22	150		150		ns
$\overline{STROBE}$ ↑ to clock ↓ setup time (to activate READY on next clock cycle)	t_{SHL7}		"	23	220		220		ns
$\overline{IORQ}$ ↑ to PORT DATA stable delay (mode 0)	t_{PZL3} t_{PZH3}		"	24		180		200	ns
PORT DATA TO $\overline{STROBE}$ ↑ setup time (mode 1)	t_{SLH8} t_{SHL8}		"	25	230		260		ns
$\overline{STROBE}$ ↓ to PORT DATA stable (mode 2)	t_{PZL4} t_{PZH4}		"	26		210		230	ns
$\overline{STROBE}$ ↑ to PORT DATA float delay (mode 2)	t_{PLZ2} t_{PHZ2}		"	27		180		200	ns
PORT DATA match to $\overline{INT}$ ↓ delay (mode 3)	t_{PHL6}		"	28		490		540	ns
$\overline{STROBE}$ ↑ to $\overline{INT}$ ↓ delay	t_{PHL7}		"	29		440		490	ns

1/ $t_{cyc} = t_{PWH1} + t_{rc} + t_{PWL1} + t_{fc}$.

2/ t_{SHL1} may be reduced, however, the time subtracted must be added to t_{PZL1} or t_{PZH1}.

3/ M1 must be active for a minimum of 2 clock cycles to reset the PIO.

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and on figure 6, and unless otherwise specified, apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements shall be as specified in table II. The subgroups of table III along with table IV and figure 6 which constitute the minimum electrical test requirements for screening and quality conformance by device class are specified in table II.

3.6 Register definition. The register definitions shall be as specified in table V.

3.7 Marking. Marking shall be in accordance with MIL-M-38510. The "JAN" or "J" certification mark shall not be used.

3.8 Microcircuit group assignment. The device covered by this specification shall be in microcircuit group 107 (see MIL-M-38510, appendix E).

3.9 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have manufacturer certification in accordance with MIL-M-38510 for at least one line. Not necessarily the line producing the device type described herein.

3.10 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III unless otherwise specified)
	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1,7
Final electrical test parameters (method 5004)	*1,2,3,*7,8, 9,10,11
Group A test requirements (method 5005)	1,2,3,7,8,9, 10,11,12
Group B V_{ZAP} test	See 4.5.2
Group C end-point electrical parameters (method 5005)	1,2,3,7,8
Group D end-point electrical parameters (method 5005)	1,2,3,7,8

*PDA applies to subgroups 1 and 7, (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3 or equivalent.

(2) T_A = 125°C, minimum.

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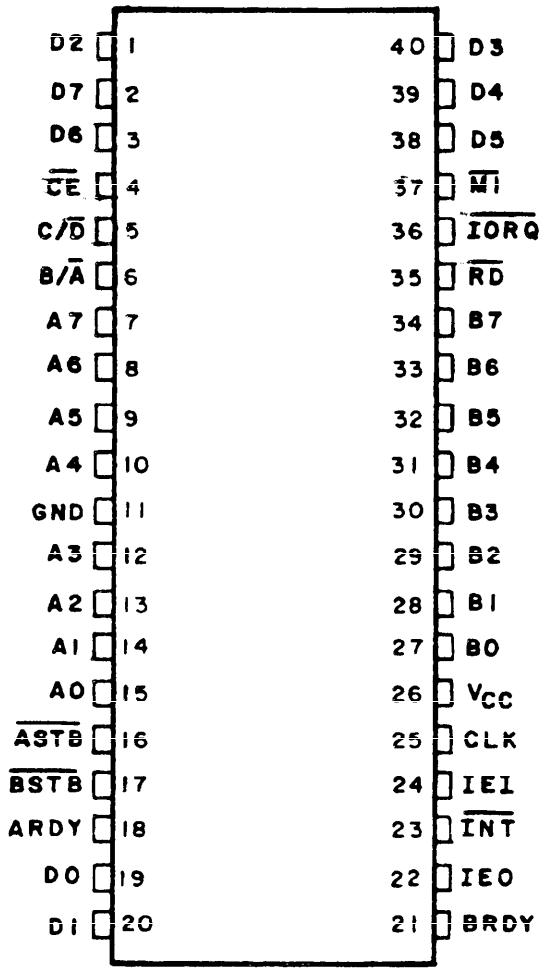


FIGURE 1. Terminal connections.

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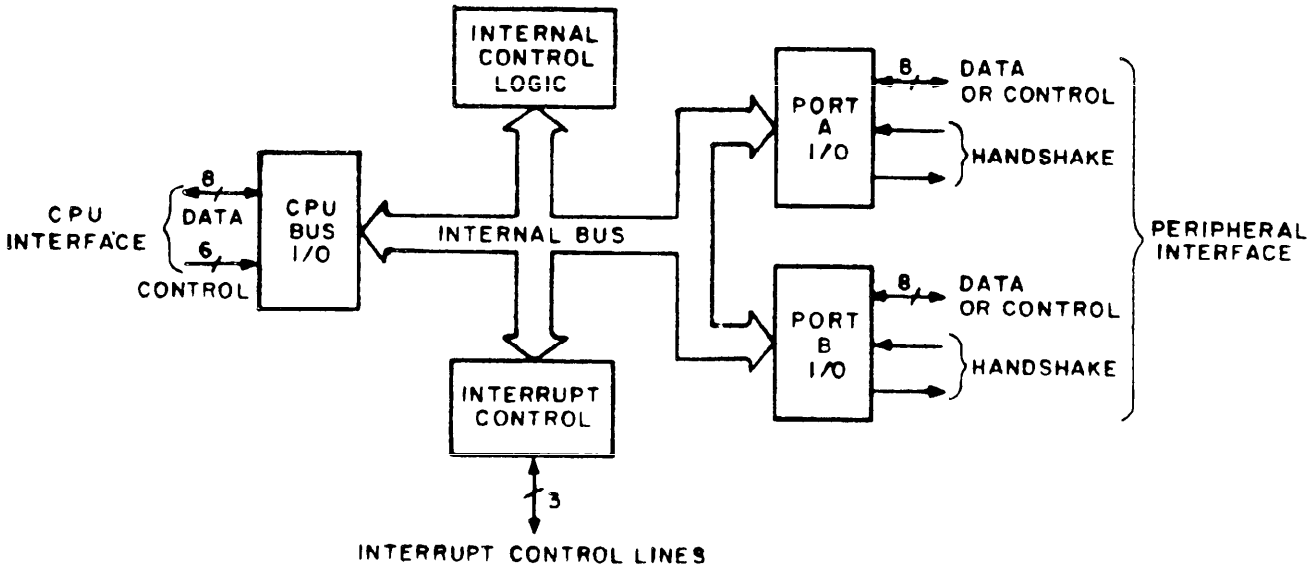
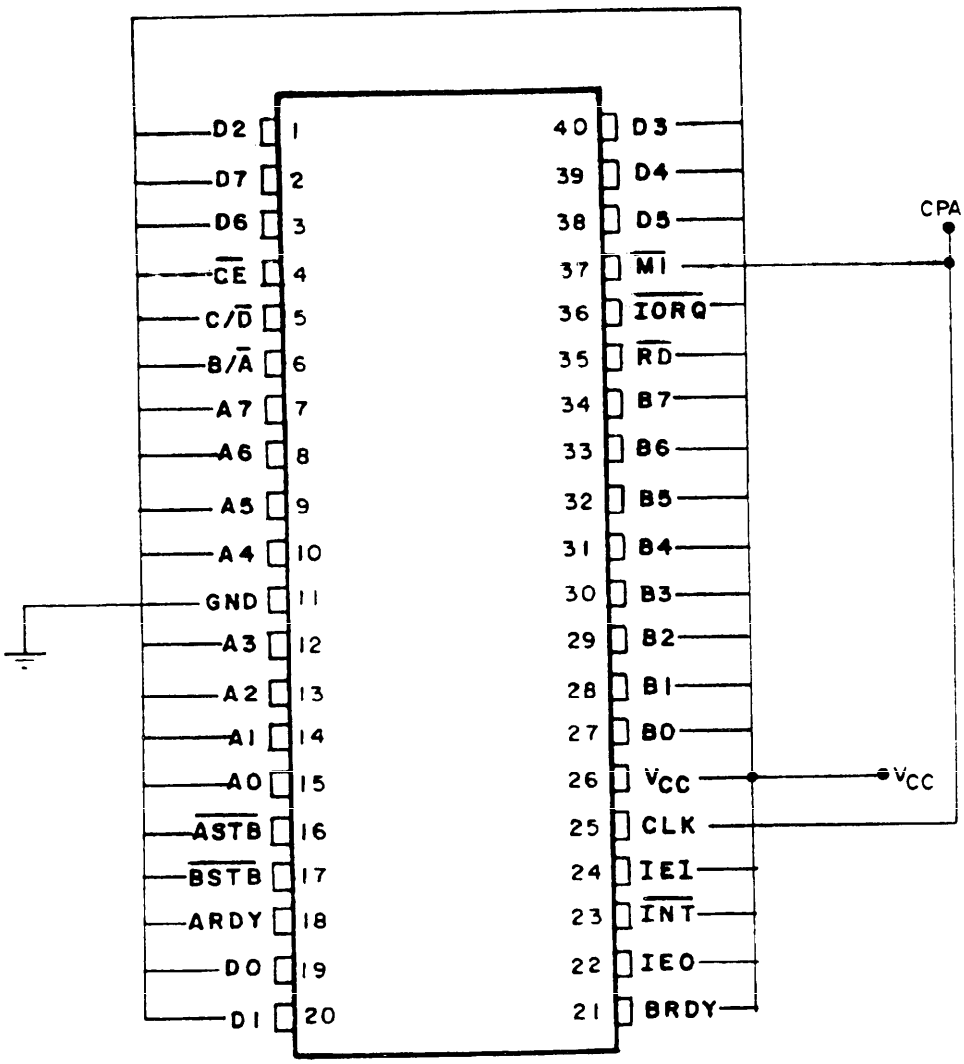


FIGURE 2. Functional block diagram.

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NOTES:
1. CPA = 100 kHz ±10% square wave.
   50% duty cycle
2. V_{CC} shall be no less than 4.5 V at device terminal.
   V_{CC} shall be no more than 5.5 V at device terminal.
3. V_{ILC} shall be no more than 0.5 V at CPA.
   V_{IHC} shall be no less than 3.9 V at CPA.

FIGURE 3. Burn-in and life test circuit.

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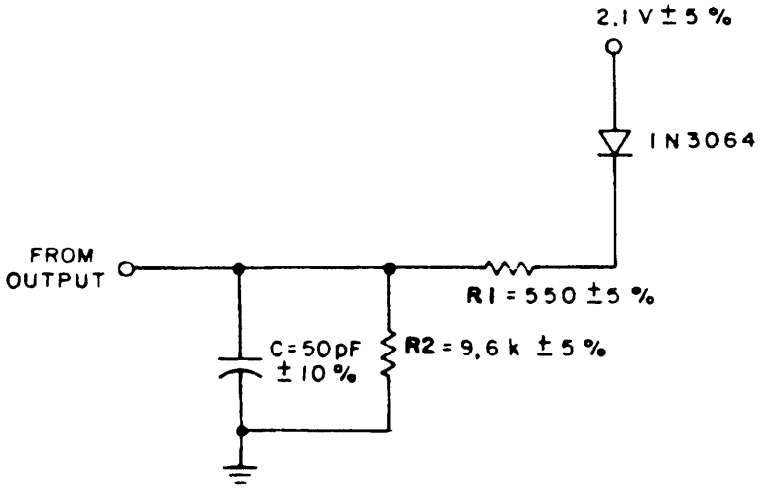
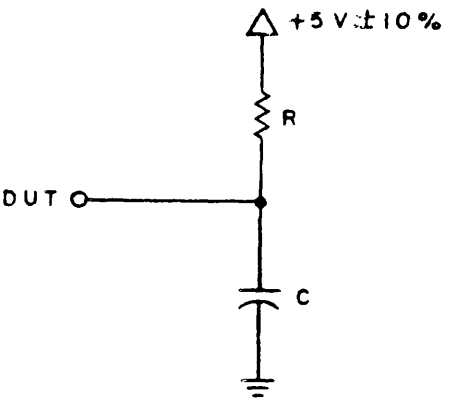


FIGURE 4. Output load circuit for ac and functional testing (standard outputs).

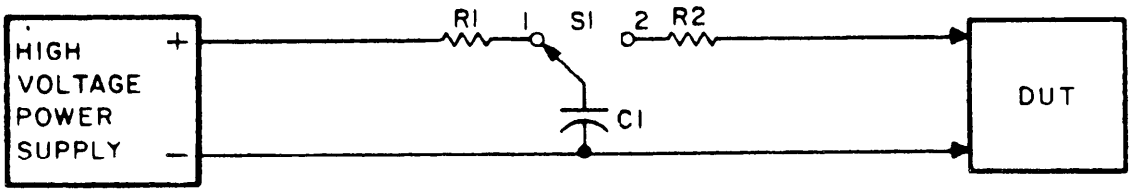
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NOTES:
1. C = 50 pF ±5%, including tester, wiring, and stray capacitance.
2. R = 2.2 kΩ ±5%.

FIGURE 4. Output load circuit for ac and functional testing (open drain outputs).

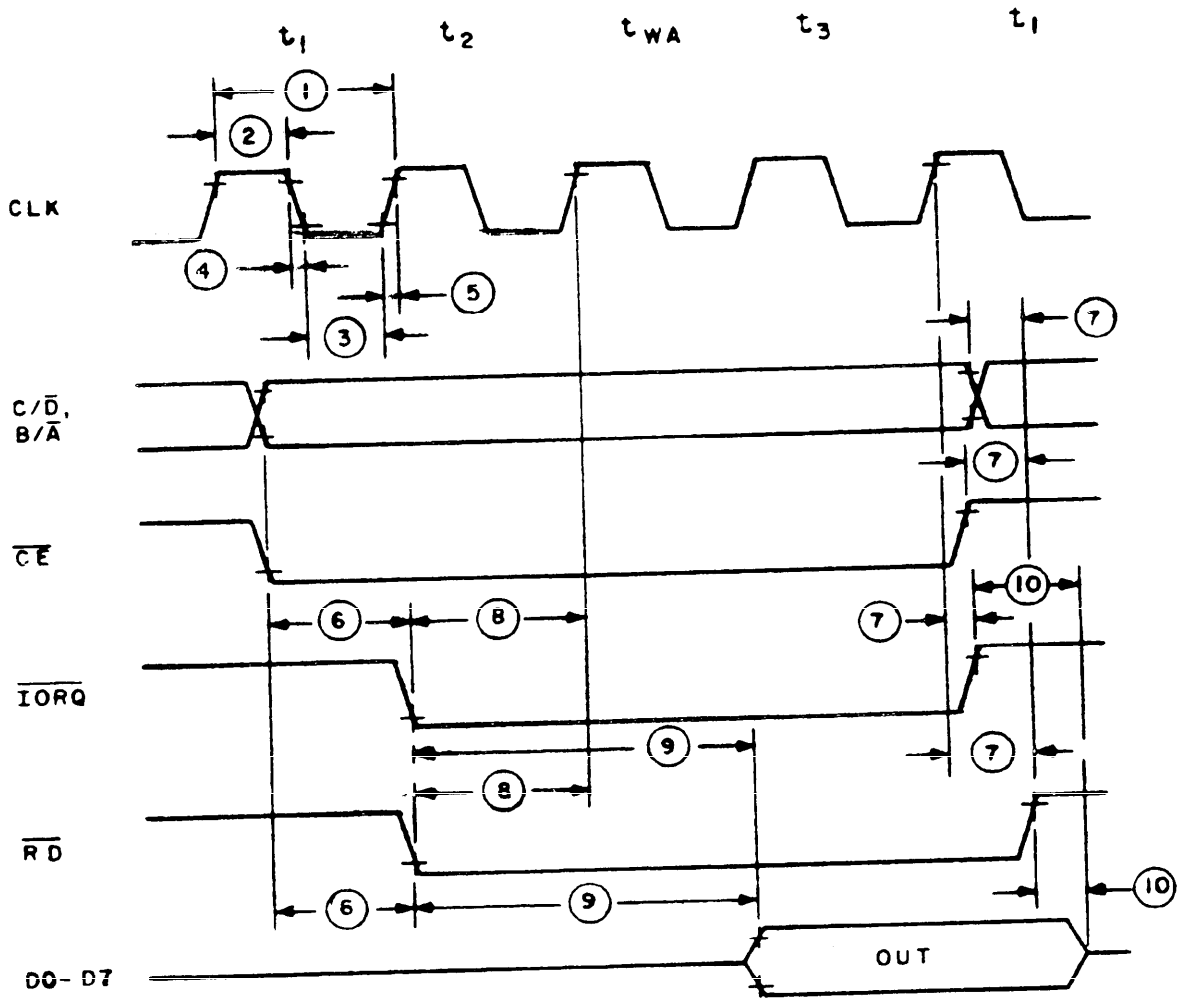
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NOTES:
1. V_{ZAP} = 400 V measured at the device input terminal.
2. 1 MΩ ≤ R1 ≤ 5 MΩ.
3. R2 = 1.5 kΩ ±5%.
4. C1 = 100 pF ±10%.
5. S1 = Hg-wetted "bounceless" relay break before make type.
6. All pins of D.U.T. not associated with V_{ZAP} test shall be open (nc).

FIGURE 5. High voltage (V_{ZAP}) test circuit.

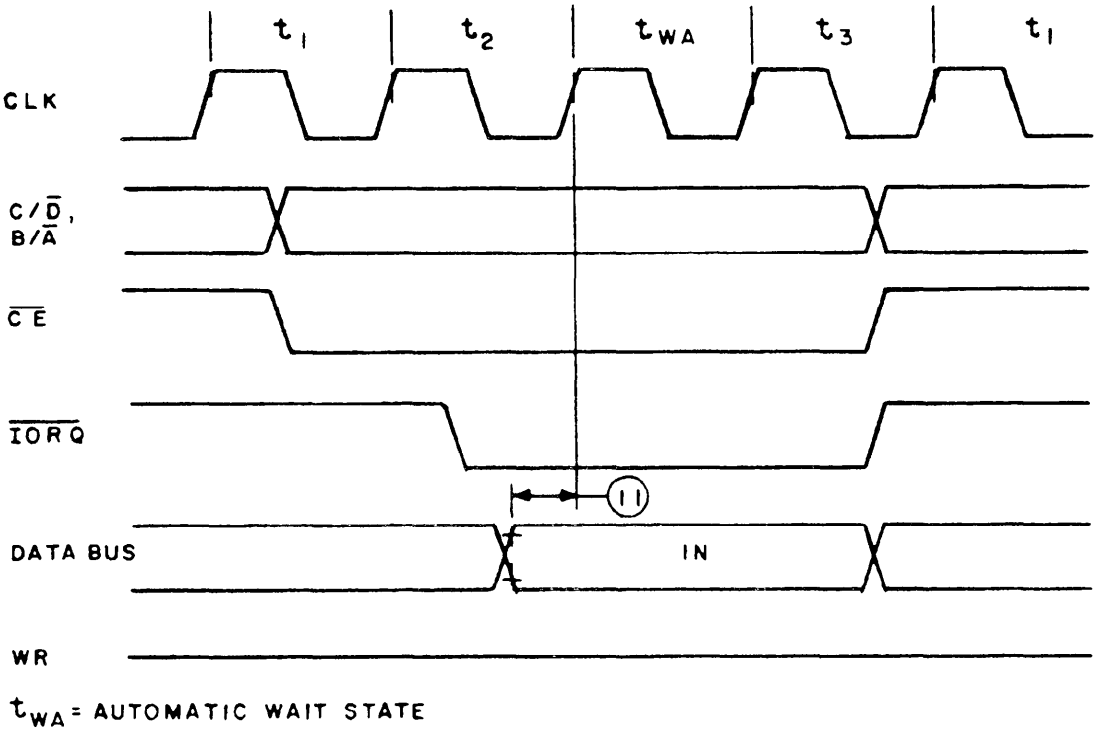
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READ CYCLE TIMING

FIGURE 6. Timing diagram.

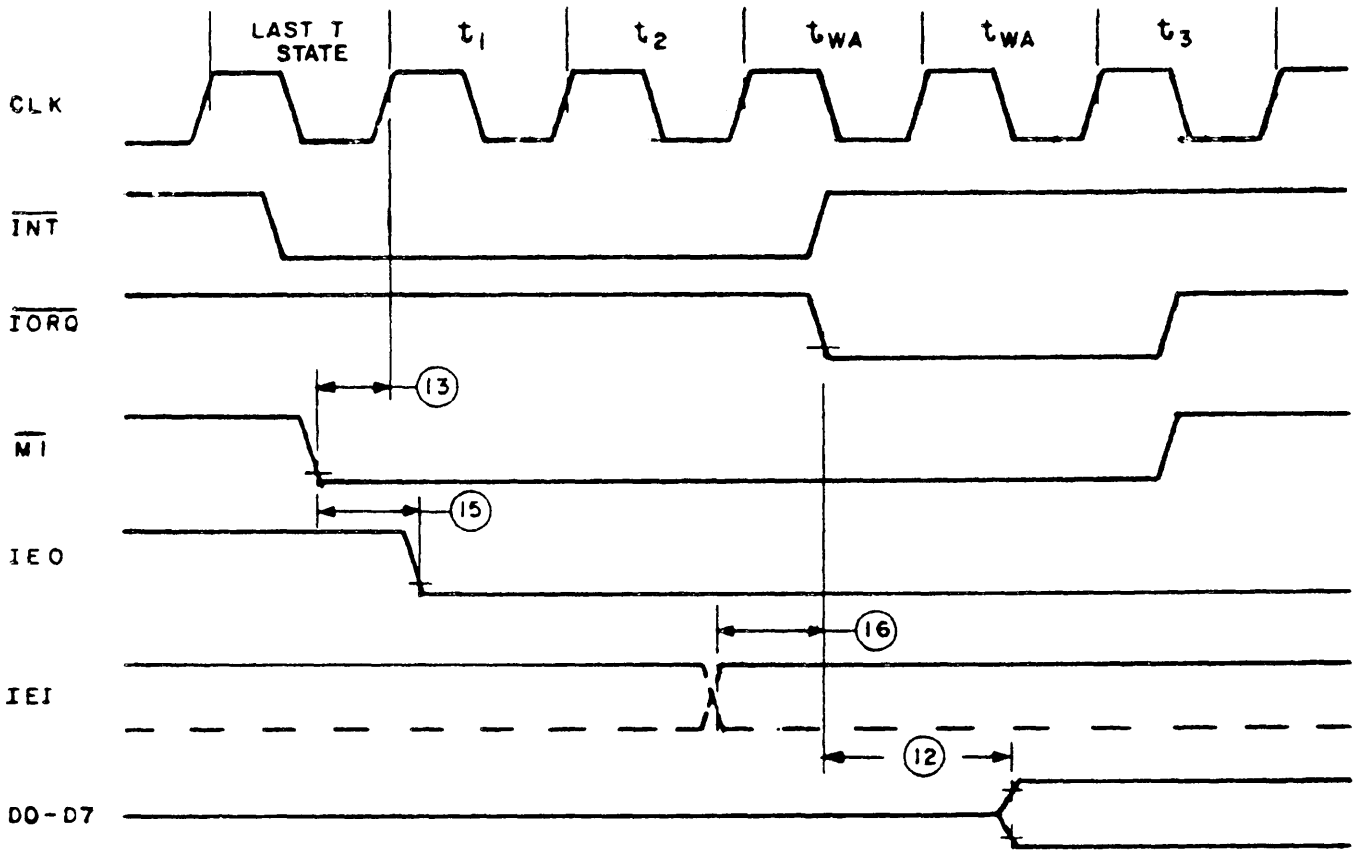
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WRITE CYCLE TIMING

FIGURE 6. Timing diagram - Continued.

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INTERRUPT ACKNOWLEDGE TIMING

FIGURE 6. Timing diagram - Continued.

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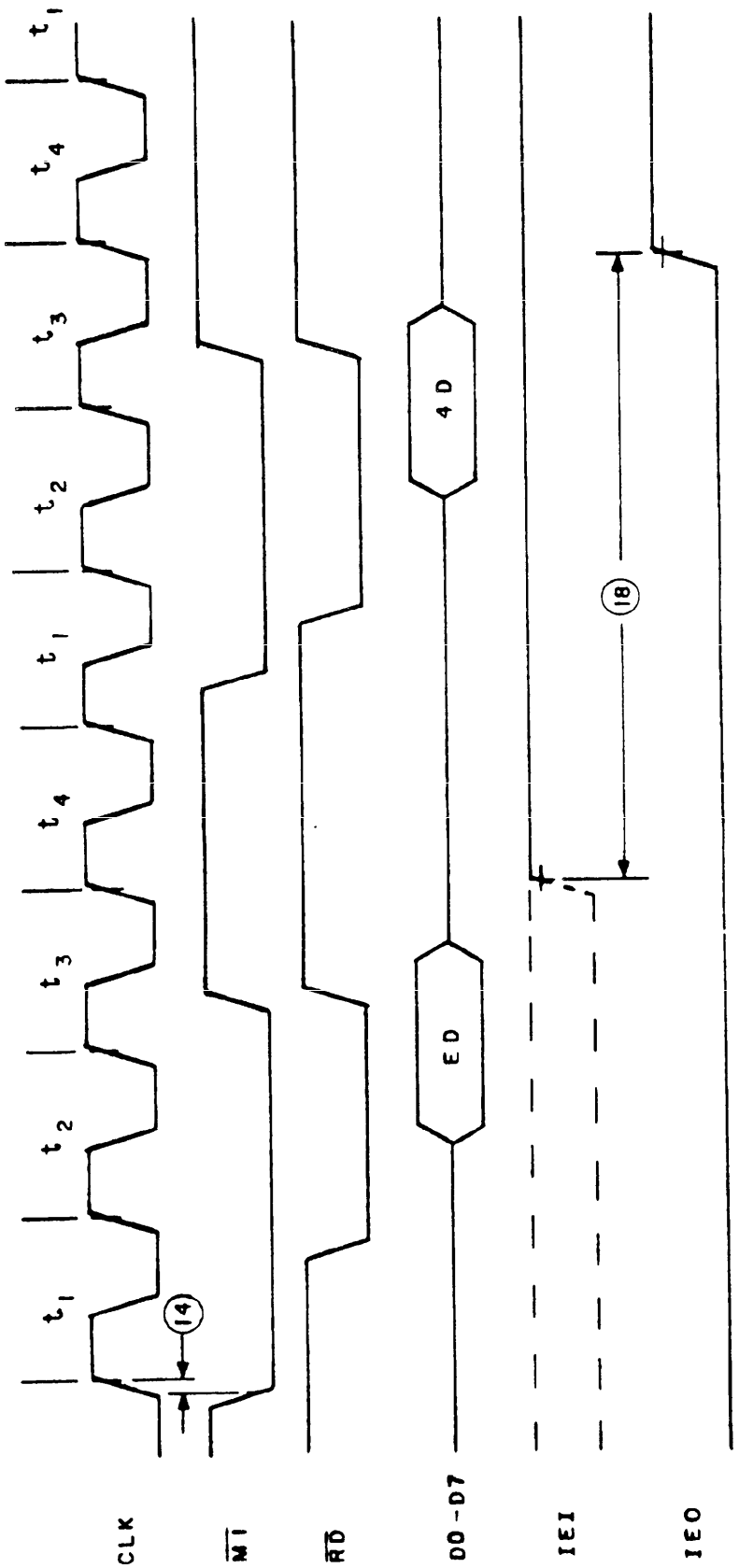
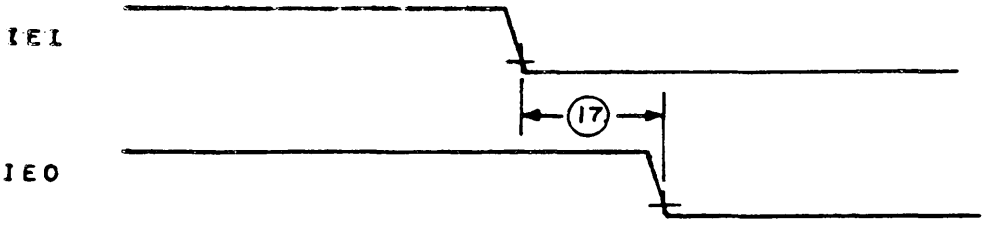


FIGURE 6. Timing diagram - Continued.

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IEI/IEO TIMING

FIGURE 6. Timing diagram - Continued.

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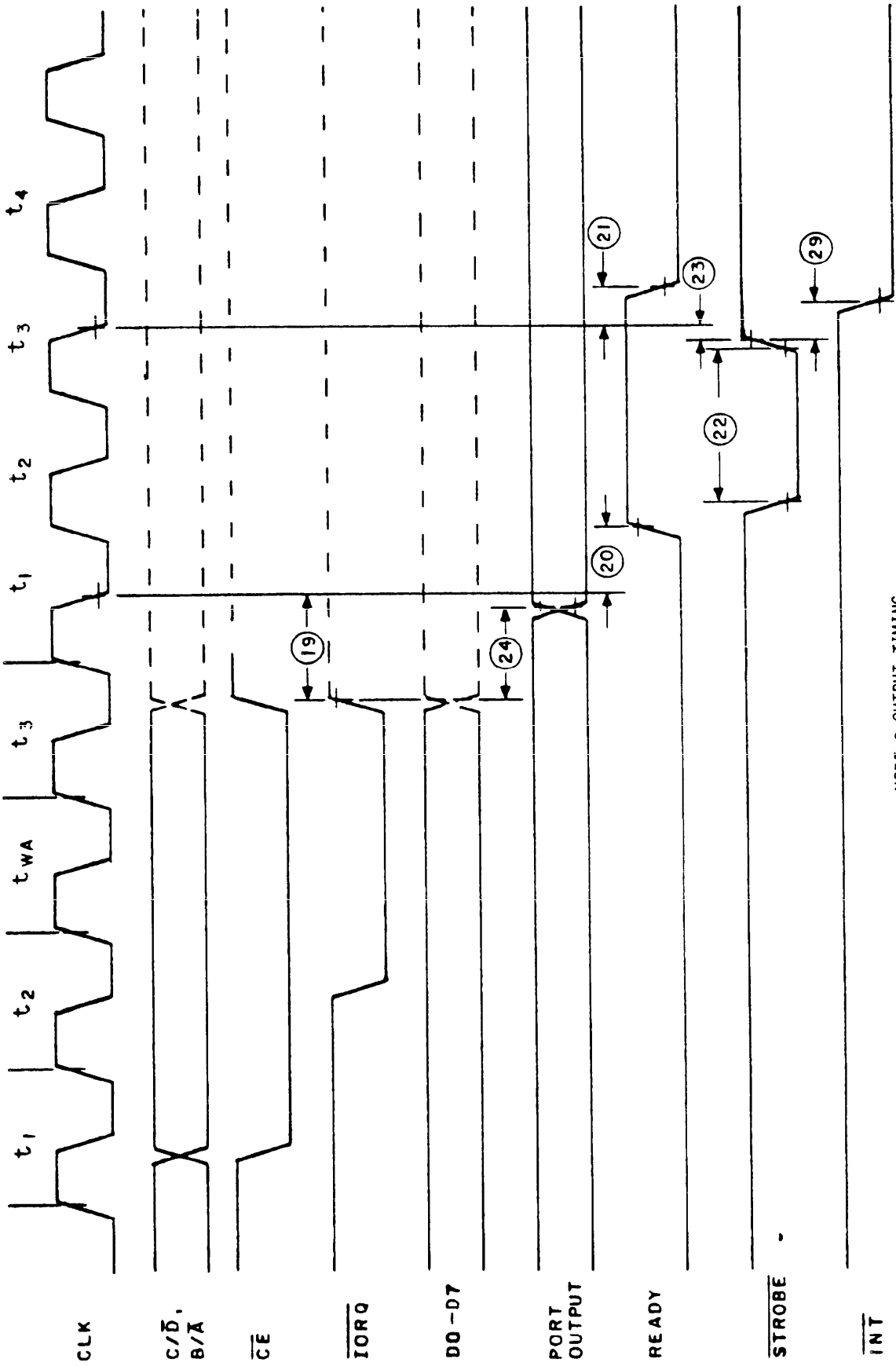
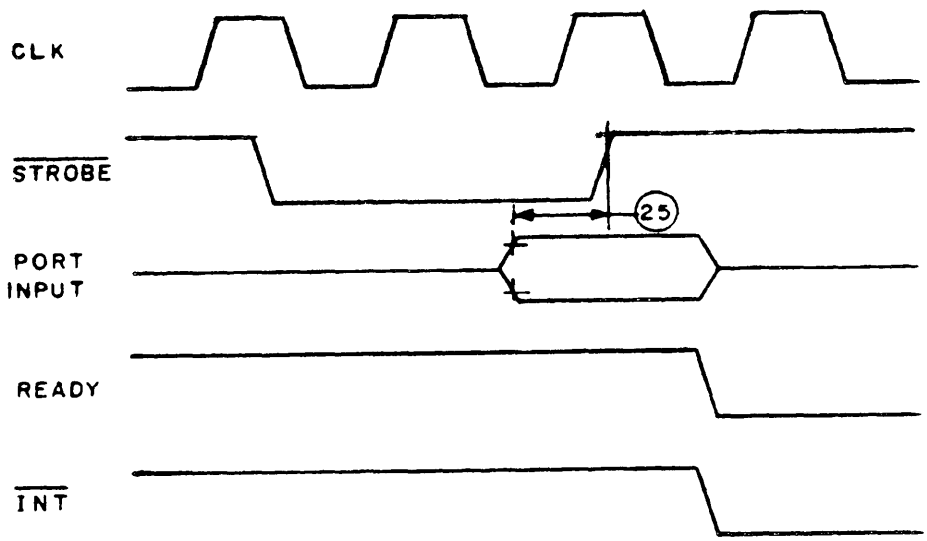


FIGURE 6. Timing diagram - Continued.

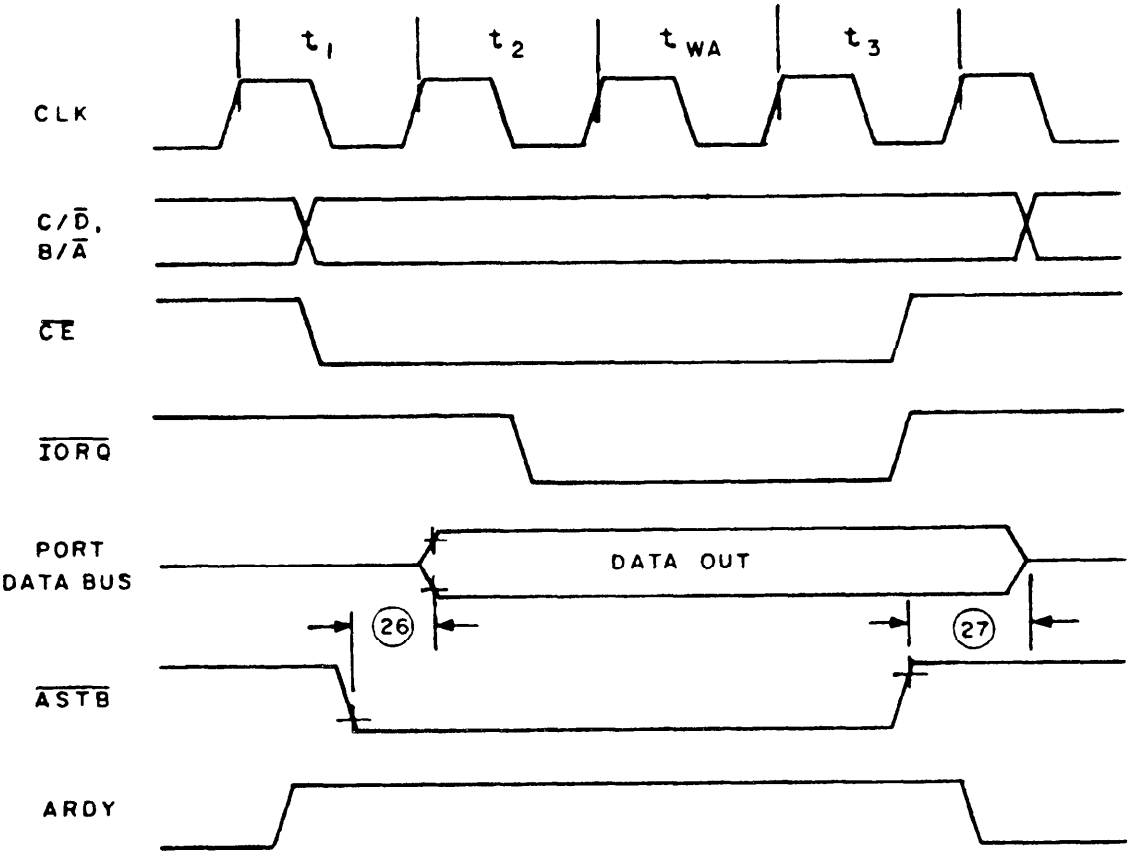
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MODE 3 INPUT TIMING

FIGURE 6. Timing diagram - Continued.

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MODE 2 BIDIRECTIONAL TIMING

WRITE DATA TO PORT A REGISTERS

FIGURE 6. Timing diagram - Continued.

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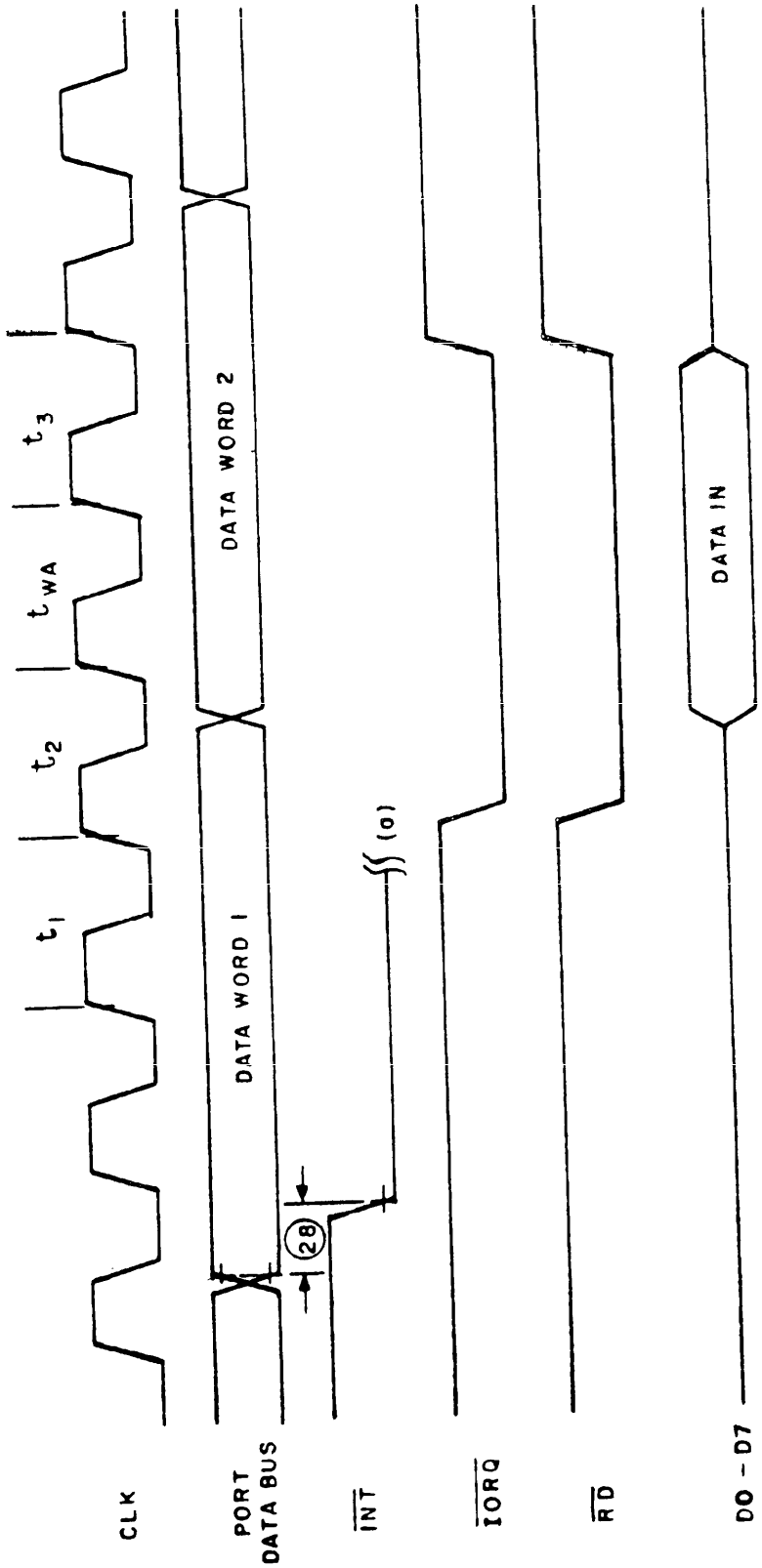


FIGURE 6. Timing diagram - Continued.

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TABLE III. Group A inspection for device types 01 and 02.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				D2	D7	D6	$\overline{CE}$	C/$\overline{D}$	B/$\overline{A}$	A7	A6	A5	A4	GND	A3	A2	A1	A0	$\overline{ASTB}$	BSTB	AKDY	D0	D1
1 T_C = +25°C	Run setup vectors 1 through 5.																						
	I_{CC}	3005	1				2.0 V							GND									
	I_{LOL}		2				"							"									
	I_{LOH}		3				"							"									
	I_{IL}		4	0.4 V			"							"									
			5		0.4 V		"							"									
			6			0.4 V	"							"									
			7				0.4 V							"									
			8				2.0 V	0.4 V						"									
			9				"		0.4 V					"									
			10				"			0.4 V				"									
			11				"				0.4 V			"									
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			43				"		2.4 V					"									
			44				"			2.4 V				"									
			45				"				2.4 V			"									
			46				"					2.4 V		"									
			47				"						2.4 V	"	2.4 V								
			48				"							"		2.4 V							
			49				"							"									

See footnotes at end of table.

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TABLE IIIC. Group A inspection for device types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Test no.	21 BRDY	22 IEO	23 $\overline{INT}$	24 IEI	25 CLK	26 V_{CC}	27 B0	28 B1	29 B2	30 B3	31 B4	32 B5	33 B6	34 B7	35 $\overline{RD}$	36 $\overline{IORQ}$	37 $\overline{M1}$	38 D5	39 D4	40 D3	Measured terminal	Test limits Min	Test limits Max	Unit
1					4.9 V	5.5 V											2.0 V				26		109	mA
2			0.4 V		"	"											"				23	-10	+10	µA
3			2.4 V		"	"											"				23	"	"	"
4																					1			
5																					2			
6																					3			
7																					4			
8																					5			
9																					6			
10																					7			
11																					8			
12																					9			
13																					10			
14																					12			
15																					13			
16																					14			
17																					15			
18																					16			
19																					17			
20																					19			
21																					20			
22				0.4 V																	24			
23					0.4 V																25			
24					4.9 V		0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V					27			
25					"																28			
26																					29			
27																					30			
28																					31			
29																					32			
30																					33			
31																					34			
32																					35			
33																					36			
34																	0.4 V	0.4 V			37			
35																	2.0 V				38			
36																	"		0.4 V		39			
37																				0.4 V	40			
38																					1			
39																					2			
40																					3			
41																					4			
42																					5			
43																					6			
44																					7			
45																					8			
46																					9			
47																					10			
48																					12			
49																					13			

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TABLE III. Group A inspection fordevice types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				02	07	06	$\overline{CE}$	C/$\overline{D}$	B/$\overline{A}$	A7	A6	A5	A4	GND	A3	A2	A1	A0	$\overline{ASTB}$	$\overline{BSTB}$	ARDY	D0	D1
1 T_C = +25°C	I_{IH4}		50				2.4 V							GND			2.4 V	2.4 V	2.4 V				
			51				"							"						2.4 V			
			52				"							"								2.4 V	
			53				"							"									2.4 V
			54				"							"									
			55				"							"									
			56				2.0 V							"									
			57				"							"									
			58				"							"									
			59				"							"									
			60				"							"									
			61				"							"									
			62				"							"									
			63				"							"									
			64				"							"									
			65				"							"									
			66				"							"									
			67				"							"									
			68				"							"									
			69				"							"									
			70				"							"									
			71				"							"									
Run setup vectors 6 through 9																							
	V_{OL}		72				2.0 V							GND								2.0 mA	
	V_{OL}		73				2.0 V							GND								2.0 mA	
Run setup vectors 10 and 11																							
	V_{OL}		74				2.0 V			2.0 mA	2.0 mA	2.0 mA	2.0 mA	GND	2.0 mA	2.0 mA	2.0 mA	2.0 mA					
			75				"			"	"	"	"	"	"	"	"	"					
			76				"			"	"	"	"	"	"	"	"	"					
			77				"			"	"	"	"	"	"	"	"	"					
			78				"			"	"	"	"	"	"	"	"	"					
			79				"			"	"	"	"	"	"	"	"	"					
			80				"			"	"	"	"	"	"	"	"	"					
			81				"			"	"	"	"	"	"	"	"	"					
			82				"			"	"	"	"	"	"	"	"	"					
			83				"			"	"	"	"	"	"	"	"	"					
			84				"			"	"	"	"	"	"	"	"	"					
			85				"			"	"	"	"	"	"	"	"	"					
			86				"			"	"	"	"	"	"	"	"	"					
			87				"			"	"	"	"	"	"	"	"	"					
			88				"			"	"	"	"	"	"	"	"	"					
			89				"			"	"	"	"	"	"	"	"	"					
Run setup vectors 12 and 13.																							
	V_{OH}		90				2.0 V			-250 µA	-250 µA	-250 µA	-250 µA	GND	-250 µA								
			91				"			"	"	"	"	"	"								
			92				"			"	"	"	"	"	"								
			93				"			"	"	"	"	"	"								
			94				"			"	"	"	"	"	"								

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TABLE IIIC. Group A inspection fordevice types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Test no.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Measured terminal	Test limits		Unit
	BRDY	IEO	$\overline{INT}$	IEI	CLK	V_{CC}	B0	B1	B2	B3	B4	B5	B6	B7	$\overline{RD}$	$\overline{TORQ}$	$\overline{HI}$	D5	D4	D3		Min	Max	
50					4.9 V	5.5 V											2.0 V				14	-10	+10	µA
51					"	"											"				15	"	"	"
52					"	"											"				16	"	"	"
53					"	"											"				17	"	"	"
54					"	"											"				19	"	"	"
55					"	"											"				20	"	"	"
56				2.4 V	"	"											"				24	"	"	"
57					5.5 V	"											"				25	"	"	"
58					4.9 V	"	2.4 V										"				27	"	"	"
59					"	"		2.4 V									"				28	"	"	"
60					"	"			2.4 V								"				29	"	"	"
61					"	"				2.4 V							"				30	"	"	"
62					"	"					2.4 V						"				31	"	"	"
63					"	"						2.4 V					"				32	"	"	"
64					"	"							2.4 V				"				33	"	"	"
65					"	"								2.4 V			"				34	"	"	"
66					"	"									2.4 V		"				35	"	"	"
67					"	"										2.4 V	"				36	"	"	"
68					"	"											2.4 V				37	"	"	"
69					"	"											2.0 V	2.4 V			38	"	"	"
70					"	"											"		2.4 V		39	"	"	"
71					"	"											"			2.4 V	40	"	"	"
72	20 mA				4.9 V	5.5 V															18		0.4	V
73					"	"															21		0.4	V
74					4.9 V	5.5 V	16 mA	16 mA	16 mA	16 mA	16 mA	16 mA	16 mA	16 mA							7		0.4	V
75					"	"															8		"	"
76					"	"															9		"	"
77					"	"															10		"	"
78					"	"															12		"	"
79					"	"															13		"	"
80					"	"															14		"	"
81					"	"															15		"	"
82					"	"															27		"	"
83					"	"															28		"	"
84					"	"															29		"	"
85					"	"															30		"	"
86					"	"															31		"	"
87					"	"															32		"	"
88					"	"															33		"	"
89					"	"															34		"	"
90					4.9 V	5.5 V															7	2.4		V
91					"	"															8	"		"
92					"	"															9	"		"
93					"	"															10	"		"
94					"	"															12	"		"

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TABLE III. Group A inspection fordevice types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				D2	D7	D6	$\overline{CE}$	C/$\overline{D}$	B/$\overline{A}$	A7	A6	A5	A4	GND	A3	A2	A1	A0	$\overline{ASTB}$	$\overline{BSTB}$	ARDY	D0	D1
1 T_C = +25°C	V_{OH}		95				2.0 V							GND		-250 µA	-250 µA	-250 µA			-250 µA		
			96				"							"									
			97				"							"									
			98				"							"									
			99				"							"									
			100				"							"									
			101				"							"									
			102				"							"									
			103				"							"									
			104				"							"									
			105				"							"									
			106				"							"									
			107				"							"									
	I_{OH2}		108				"							"									
			109				"							"									
			110				"							"									
			111				"							"									
			112				"							"									
			113				"							"									
			114				"							"									
			115				"							"									
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and the limit for test 1 = 91 mA.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and the limit for test 1 = 127 mA.																						

See footnotes at end of table.

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TABLE IIIC. Group A inspection fordevice types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Test no.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Measured terminal	Test limits		Unit
	BRDY	IEO	$\overline{INT}$	IEI	CLK	V_{CC}	B0	B1	B2	B3	B4	B5	B6	B7	$\overline{RD}$	$\overline{TURQ}$	$\overline{MI}$	D5	D4	D3		Min	Max	
95					4.9 V	5.5 V															13	2.4		V
96					"	"															14	"		"
97					"	"															15	"		"
98					"	"															18	"		"
99	-250 µA				"	"	-250 µA	-250 µA	-250 µA	-250 µA	-250 µA	-250 µA	-250 µA	-250 µA							21	"		"
100	"				"	"	"	"	"	"	"	"	"	"							27	"		"
101	"				"	"	"	"	"	"	"	"	"	"							28	"		"
102	"				"	"	"	"	"	"	"	"	"	"							29	"		"
103	"				"	"	"	"	"	"	"	"	"	"							30	"		"
104	"				"	"	"	"	"	"	"	"	"	"							31	"		"
105	"				"	"	"	"	"	"	"	"	"	"							32	"		"
106	"				"	"	"	"	"	"	"	"	"	"							33	"		"
107	"				"	"	"	"	"	"	"	"	"	"							34	"		"
108					"	"	1.5 V	1.5 V	1.5 V	1.5 V	1.5 V	1.5 V	1.5 V	1.5 V							27	-1.5	-3.8	mA
109					"	"	"	"	"	"	"	"	"	"							28	"	"	"
110					"	"	"	"	"	"	"	"	"	"							29	"	"	"
111					"	"	"	"	"	"	"	"	"	"							30	"	"	"
112					"	"	"	"	"	"	"	"	"	"							31	"	"	"
113					"	"	"	"	"	"	"	"	"	"							32	"	"	"
114					"	"	"	"	"	"	"	"	"	"							33	"	"	"
115					"	"	"	"	"	"	"	"	"	"							34	"	"	"

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TABLE III. Group A inspection - setup vectors for subgroups 1, 2, and 3.

Vector number	D7	D6	D5	D4	D3	D2	D1	D0	$\overline{CE}$	B/$\overline{A}$	C/$\overline{D}$	$\overline{RD}$	$\overline{IORQ}$	$\overline{MI}$	Clock cycles	Comment
1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	4	Reset device
2	0	1	0	0	1	1	1	1	0	0	1	1	0	1	4	Set port A to input mode (mode 1)
3	1	1	1	1	1	1	1	0	0	0	1	1	0	1	4	Port A interrupt vector
4	0	1	0	0	1	1	1	1	0	1	1	1	0	1	4	Set port B to input mode (mode 1)
5	1	1	1	1	1	1	0	0	0	1	1	1	0	1	4	Port B interrupt vector
6	0	0	0	0	1	1	1	1	0	0	1	1	0	1	4	Set port A to output mode (mode 0)
7	1	1	1	1	1	1	1	0	0	0	1	1	0	1	4	Port A interrupt vector
8	0	0	0	0	1	1	1	1	0	1	1	1	0	1	4	Set port B to output mode (mode 0)
9	1	1	1	1	1	1	0	0	0	1	1	1	0	1	4	Port B interrupt vector
10	0	0	0	0	0	0	0	0	0	0	0	1	0	1	4	Write data to port A
11	0	0	0	0	0	0	0	0	0	1	0	1	0	1	4	Write data to port B
12	1	1	1	1	1	1	1	1	0	0	0	1	0	1	4	Write data to port A
13	1	1	1	1	1	1	1	1	0	1	0	1	0	1	4	Write data to port B

NOTE: After setting up the specified levels on the inputs (1 = 2.0 V, 0 = 0.8 V), pulse the clock input the required number of times.

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TABLE III. Group A inspection fordevice types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				D2	D7	D6	$\overline{CE}$	C/$\overline{D}$	B/$\overline{A}$	A7	A6	A5	A4	GND	A3	A2	A1	A0	$\overline{ASTB}$	$\overline{BSTB}$	ARDY	D0	D1
4 T_C = +25°C	C_{11}		116	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND
	C_{12}		117 118 119 120 121 122 123 124 125	"	"	"	1/ GND	"	1/ GND	"	"	"	"	"	"	"	"	"	1/ GND	1/ GND	"	"	"
	C_0		126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153	1/ GND	1/ GND	1/ GND	"	"	"	1/ GND	1/ GND	1/ GND	1/ GND	"	"	1/ GND	1/ GND	1/ GND	1/ GND	"	"	1/ GND	1/ GND	1/ GND
7 T_C = +25°C	2/3/5/																						
8	2/3/5/	Same tests, terminal conditions, and limits as subgroup 7, except T_C = +125°C ad test 1 limit = 91 mA and T_C = -55°C with test 1 limit = 127 mA.																					
9	2/3/4/																						
10	2/3/4/																						
11	2/3/4/																						

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TABLE IIIC. Group A inspection fordevice types 01 and 02 - Continued.
Terminal conditions (pins not designated are open)

Test no.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Measured terminal	Test limits		Unit
	BRDY	IEO	$\overline{INT}$	IEI	CLK	V_{CC}	B0	B1	B2	B3	B4	B5	B6	B7	$\overline{RD}$	$\overline{IORQ}$	$\overline{MI}$	D5	D4	D3		Min	Max	
116	GND	GND	GND	GND	1/	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	25		20	pF
117	"	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	4		3	"
118	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	5		"	"
119	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	6		"	"
120	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	16		"	"
121	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	17		"	"
122	"	"	"	1/	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	24		"	"
123	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	1/	"	"	"	"	"	35		"	"
124	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	1/	"	"	"	"	36		"	"
125	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	1/	"	"	"	37		"	"
126	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	1		10	"
127	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	2		"	"
128	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	3		"	"
129	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	7		"	"
130	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	8		"	"
131	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	9		"	"
132	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	10		"	"
133	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	12		"	"
134	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	13		"	"
135	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	14		"	"
136	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	15		"	"
137	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	18		"	"
138	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	19		"	"
139	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	20		"	"
140	1/	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	21		"	"
141	GND	1/	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	22		"	"
142	"	GND	1/	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	23		"	"
143	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	27		"	"
144	"	"	"	"	"	"	"	1/	"	"	"	"	"	"	"	"	"	"	"	"	"	"	28		"	"
145	"	"	"	"	"	"	"	GND	1/	"	"	"	"	"	"	"	"	"	"	"	"	"	29		"	"
146	"	"	"	"	"	"	"	"	GND	1/	"	"	"	"	"	"	"	"	"	"	"	"	30		"	"
147	"	"	"	"	"	"	"	"	"	GND	1/	"	"	"	"	"	"	"	"	"	"	"	31		"	"
148	"	"	"	"	"	"	"	"	"	"	GND	1/	"	"	"	"	"	"	"	"	"	"	32		"	"
149	"	"	"	"	"	"	"	"	"	"	"	GND	1/	"	"	"	"	"	"	"	"	"	33		"	"
150	"	"	"	"	"	"	"	"	"	"	"	"	GND	1/	"	"	"	"	"	"	"	"	34		"	"
151	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	1/	"	"	"	"	"	"	"	38		"	"
152	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	1/	"	"	"	"	"	"	39		"	"
153	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	"	"	"	40		"	"

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TABLE III. Group A inspection - Continued.

Test number	Symbol	Terminal(s)	Figure number	Reference number	Subgroup 12 limits +25°C, +125°C				Unit
					Device 01		Device 02		
					Min	Max	Min	Max	
154	t_{PZL1} t_{PZH1}	D0-D7	6	9		380		430	ns
155	t_{PLZ1} t_{PHZ1}	D0-D7	"	10		110		160	"
156	t_{PZL2} t_{PZH2}	D0-D7	"	12	250		340		"
157	t_{PHL1}	$\overline{MI}$	"	15		190		300	"
158	t_{PHL2}	IE0	"	17		130		190	"
159	t_{PLH3}	IE0	"	18		160		210	"
160	t_{PHL4}	Ready	"	20	190		200		"
161	t_{PHL5}	Ready	"	21	140		150		"
162	t_{PZL3} t_{PZH3}	Port data bus	"	24		180		250	"
163	t_{PZL4} t_{PZH4}	Port data bus	"	26		210		230	"
164	t_{PLZ2} t_{PHZ2}	Port data bus	"	27		180		200	"
165	t_{PLH6}	INT	"	28		490		540	"
166	t_{PHL7}	INT	"	29		440		490	"

See footnotes at end of table.

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1/ The capacitance measurement shall be made between the indicated terminal and ground at a frequency of 1 MHz. The dc bias of the measurement instrument shall be less than ±0.1 V. The ac amplitude shall be less than 500 mV rms.

2/ For functional tests, the input/output conditions A, B, C, And D appear in table IV.

3/ Table VI functional tests shall be repeated for input/output conditions A through D of table IV.

4/ All subgroups 9, 10, and 11 are run concurrently with subgroups 7 and 8.

5/ V_{OH} and V_{OL} measurements not made as part of subgroups 1 through 3 are measured as a part of subgroups 7 and 8.

6/ Where propagation delay tests for subgroup 12 indicate a group of pins and where two or more propagation delays are specified for a test, only the reading for the pin with the worst delay requires recording. In some cases, this will be a minimum reading, while in others it will be a maximum reading, as indicated in the table.

7/ Table VI functional tests shall be repeated for input/output conditions A through D of table IV for tests 154 through 166.

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TABLE IV. Input/output conditions for table III, device type 01.

Symbol	Reference number	Terminal(s)	Input/output conditions				Unit
			A	B	C	D	
V_{CC}	---	V_{CC}	5.5	4.5	4.5	4.5	V
V_{IH1}	---	CLK	4.9	3.9	3.9	3.9	V
V_{IL1}	---	CLK	0.45	0.45	0.45	0.45	V
V_{IH2}	---	Logic inputs	2.0	2.0	2.0	2.0	V
V_{IL2}	---	Logic inputs	0.8	0.8	0.8	0.8	V
V_{OH}	---	Logic outputs	2.4	2.4	2.4	2.4	V
V_{OL}	---	Logic outputs	0.4	0.4	0.4	0.4	V
t_{cyc1}	1	CLK	250	250	2000	2000	ns
t_{PWH1}	2	CLK	105	105	105		ns
t_{PWL1}	3	CLK				105	ns
t_{FC1}	4	CLK	20	20	20	20	ns
t_{RC1}	5	CLK	20	20	20	20	ns
t_{SHL1}	6	$\overline{CE}$	50	50	50	50	ns
t_{HHL} t_{HLH}	7	Logic inputs	0	0	0	0	ns
t_{SHL2} t_{SLH2}	8	$\overline{RD}$	115	115	115	115	ns
t_{PZL1} t_{PZH1}	9	$\overline{RD}$	380	380	380	380	ns
t_{PLZ1} t_{PHZ1}	10	$\overline{RD}$	110	110	110	110	ns
t_{SZH1} t_{SZL1}	11	D0-D7	50	50	50	50	ns
t_{PZL2} t_{PZH2}	12	$\overline{IORQ}$	160	160	160	160	ns
t_{SHL3}	13	$\overline{M1}$	90	90	90	90	ns
t_{SLH4}	14	$\overline{M1}$	0	0	0	0	ns
t_{PHL1}	15	$\overline{M1}$	190	190	190	190	ns
t_{SHL5}	16	$\overline{IORQ}$	140	140	140	140	ns
t_{PHL2}	17	IEI	130	130	130	130	ns
t_{PLH3}	18	IEI	160	160	160	160	ns
t_{SLH6}	19	$\overline{IORQ}$	200	200	200	200	ns
t_{PHL4}	20	Ready	190	190	190	190	ns

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TABLE IV. Input/output conditions for table III, device type 01 - Continued.

Symbol	Reference number	Terminal(s)	Input/output conditions A	B	C	D	Unit
t_{PHL5}	21	Ready	140	140	140	140	ns
t_{PWL2}	22	Strobe	150	150	150	150	ns
t_{SHL7}	23	Strobe	220	220	220	220	ns
t_{PZL3} t_{PZH3}	24	$\overline{IORQ}$	180	180	180	180	ns
t_{SLH8} t_{SHL8}	25	Port data bus	230	230	230	230	ns
t_{PZL4} t_{PZH4}	26	Port data bus	210	210	210	210	ns
t_{PLZ2} t_{PHZ2}	27	Port data bus	180	180	180	180	ns
t_{PHL6}	28	$\overline{INT}$	490	490	490	490	ns
t_{PHL7}	29	$\overline{INT}$	440	440	440	440	ns

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TABLE IV. Input/output conditions for table III, device type 02.

Symbol	Reference number	Terminal(s)	Input/output conditions				Unit
			A	B	C	D	
V_{CC}	---	V_{CC}	5.5	4.5	4.5	4.5	V
V_{IH1}	---	CLK	4.9	3.9	3.9	3.9	V
V_{IL1}	---	CLK	0.45	0.45	0.45	0.45	V
V_{IH2}	---	Logic inputs	2.0	2.0	2.0	2.0	V
V_{IL2}	---	Logic inputs	0.8	0.8	0.8	0.8	V
V_{OH}	---	Logic outputs	2.4	2.4	2.4	2.4	V
V_{OL}	---	Logic outputs	0.4	0.4	0.4	0.4	V
t_{cyc1}	1	CLK	400	400	2000	2000	ns
t_{PWH1}	2	CLK	170	170	170		ns
t_{PWL1}	3	CLK				170	ns
t_{FC1}	4	CLK	30	30	30	30	ns
t_{RC1}	5	CLK	30	30	30	30	ns
t_{SHL1}	6	$\overline{CE}$	50	50	50	50	ns
t_{HHL} t_{HLH}	7	Logic inputs	0	0	0	0	ns
t_{SHL2} t_{SLH2}	8	$\overline{RD}$	115	115	115	115	ns
t_{PZL1} t_{PZH1}	9	$\overline{RD}$	430	430	430	430	ns
t_{PLZ1} t_{PHZ1}	10	$\overline{RD}$	160	160	160	160	ns
t_{SZH1} t_{SZL1}	11	D0-D7	50	50	50	50	ns
t_{PZL2} t_{PZH2}	12	$\overline{IORQ}$	340	340	340	340	ns
t_{SHL3}	13	$\overline{M1}$	210	210	210	210	ns
t_{SLH4}	14	$\overline{M1}$	0	0	0	0	ns
t_{PHL1}	15	$\overline{M1}$	300	300	300	300	ns
t_{SHL5}	16	$\overline{IORQ}$	140	140	140	140	ns
t_{PHL2}	17	IEI	190	190	190	190	ns
t_{PLH3}	18	IEI	210	210	210	210	ns
t_{SLH6}	19	$\overline{IORQ}$	220	220	220	220	ns
t_{PHL4}	20	Ready	200	200	200	200	ns

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TABLE IV. Input/output conditions for table III, device type 02- Continued.

Symbol	Reference number	Terminal(s)	Input/output conditions				Unit
			A	B	C	D	
t_{PHL5}	21	Ready	150	150	150	150	ns
t_{PWL2}	22	Strobe	150	150	150	150	ns
t_{SHL7}	23	Strobe	220	220	220	220	ns
t_{PZL3} t_{PZH3}	24	$\overline{IORQ}$	200	200	200	200	ns
t_{SLH8} t_{SHL8}	25	Port data bus	260	260	260	260	ns
t_{PZL4} t_{PZH4}	26	Port data bus	230	230	230	230	ns
t_{PLZ2} t_{PHZ2}	27	Port data bus	200	200	200	200	ns
t_{PHL6}	28	$\overline{INT}$	540	540	540	540	ns
t_{PHL7}	29	$\overline{INT}$	490	490	490	490	ns

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TABLE V. Register definitions.

PROGRAMMING:

Mode 0, 1, or 2 (byte input, output, or bidirectional). Programming a port for mode 0, 1, or 2 requires two words per port. These words are:

A mode control word. Selects the port operating mode (figure 7). This word may be written any time.

An interrupt vector. The Z-80 PIO is designed for use with the Z-80 CPU in interrupt mode 2 (figure 8). When interrupts are enabled, the PIO must provide an interrupt vector.

Mode 3 (bit input/output). Programming a port for Mode 3 operation requires a control word, a vector (if interrupts are enabled), and three additional words, described as follows:

I/O register control. When mode 3 is selected, the mode control word must be followed by another control word that sets the I/O control register, which in turn defines which port lines are inputs and which are outputs (figure 9).

Interrupt control word. In mode 3, handshake is not used. Interrupts are generated as a logic function of the input signal levels. The interrupt control word sets the logic conditions and the logic levels required for generating an interrupt. Two logic conditions or functions are available: AND (if all input bits change to the active level, an interrupt is triggered), and OR (if any one of the input bits changes to the active level, an interrupt is triggered). Bit D6 sets the logic function, as shown on figure 10. The active level of the input bits can be set either High or Low. The active level is controlled by bit D5.

Mask control word. This word sets the mask control register, allowing any unused bits to be masked off. If any bits are to be masked, then D4 must be set. When D4 is set, the next word written to the port must be a mask control word (figure 11).

Interrupt disable. There is one other control word which can be used to enable or disable a port interrupt. It can be used without changing the rest of the interrupt control word (figure 12).

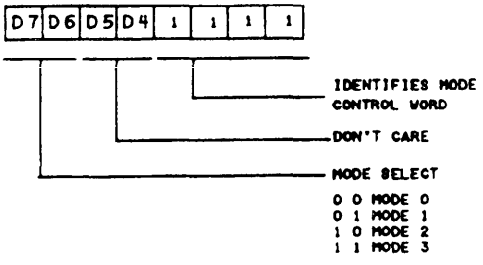


FIGURE 7. Mode control word.

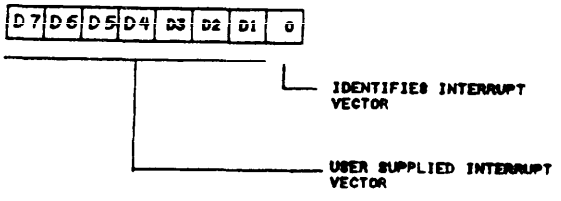


FIGURE 8. Interrupt vector word.

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TABLE V. Register definitions - Continued.

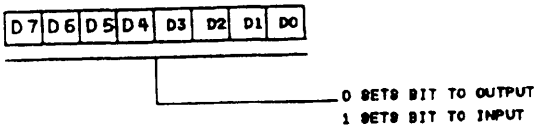
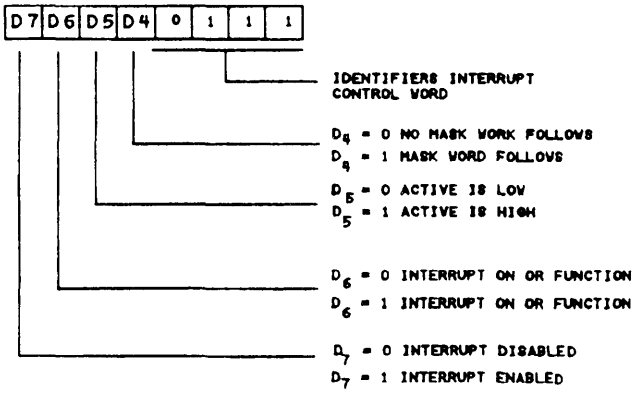


FIGURE 9. I/O register control word.



NOTE: The port is not enabled until the interrupt enable is followed by an active $\overline{M1}$.

FIGURE 10. Interrupt control word.

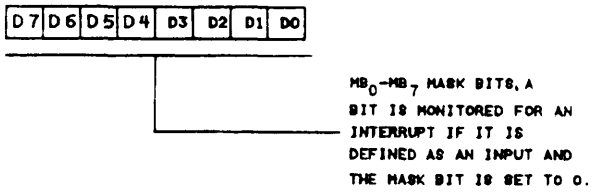


FIGURE 11. Mask control word.

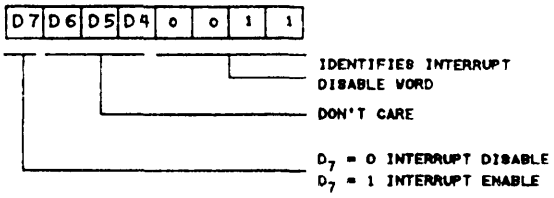


FIGURE 12. Interrupt disable word.

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b. Interim and final electrical test parameters shall be as specified in table II herein, except interim electrical parameter tests prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) is specified as 10 percent for class B based on failures from group A, subgroups 1 and 7 tests after cooldown from burn-in as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omittted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroups 1 and 7 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for the lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

d. Constant acceleration in accordance with test condition D of method 2001 of MIL-STD-883.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.3) Generic test data (see 6.8) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroup 4 (C_{I1}, C_{I2}, and C_O measurements) shall be performed only after process or design changes which may affect device capacitance.

c. Subgroup 12 shall be added to group A inspection requirements for class B devices using an LTPD of 10 and consists of the procedures, test conditions, and limits specified in table III. Variables data shall be taken for subgroup 12 only during initial verification and after process or design changes.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of method 5005 of MIL-STD-883. A special subgroup shall be added to the group B inspection requirements for class B. This subgroup shall consist of the tests, conditions, and limits specified in 4.5.2. The LTPD for this subgroup shall be 15.

4.4.3 Groups C and D inspections. Groups C and D inspections shall consist of the test subgroups and LTPD values shown in tables III and IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II.

b. Steady-state life test (method 1005 of MIL-STD-883).

(1) Test condition D or E using the circuit shown on figure 3.

(2) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Constant acceleration in accordance with test condition D of method 2001 of MIL-STD-883.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate table and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 High voltage (V_{ZAP}) test of input protection circuits. One input terminal of the device under test (DUT) shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit on figure 5.

a. Measure I_{IH} and I_{IL} at one input terminal of the device under test at 25°C. These measurements shall be made in accordance with table III herein. The test limits for a single terminal measurement of I_{IH} and I_{IL} shall be as specified in table III.

b. Apply the test voltage (V_{ZAP} = 400 V) to the same terminal selected for leakage current measurements. Apply V_{ZAP} in a 2 pulse sequence as follows and in accordance with the test circuit on figure 5.

(1) Input (-) to ground.

(2) Input (+) to ground.

c. Within 24 hours repeat the I_{IH} and I_{IL} measurements on the same terminal as performed above. After the V_{ZAP} test, a DUT exhibiting leakage currents in excess of the specified limits is defective.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Test vector table. Table VI, which forms a part of this specification, is not printed herein because of its extreme length and complexity; and, even when presented as hard copy, it requires conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be obtained in a form compatible with test system architecture (magnetic tape usually preferred). The preparing activity, or Defense Electronics Supply Center as its agent, may be consulted for information on obtaining such output. For reference purposes, Rome Air Development Center (RADC-RBE-2) and Defense Electronics Supply Center (DESC-ECS) each maintain a limited quantity of dated, hard copy printouts of this table.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

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f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
f_{MAX}	Maximum clock frequency
V_{IN}	Voltage level at an input terminal

6.5 Terms and definitions. The abbreviations, terms, symbols, and definitions used herein (including terms and symbols for device terminals) are defined in MIL-M-38510, MIL-STD-1331, and as follows:

IEI — Interrupt enable in (input, ative high) When this line is active, the PIO is able to interrupt the CPU.

IEO — Interrupt enable out (output, active high). This output is high only if IEI is high and the CPU is not servicing an interrupt from this PIO. In conjunction with IEI, this line can be used to implement a system-wide interrupt priority daisy chain.

A0-A7 — Port A bus (bidirectional, tristate). This 8 bit bus is used to transfer data and/or status or control information between Port A of the PIO and a periperal device. A0 is the least significant bit.

$\overline{ASTB}$ — Port A strobe pulse (input, active low). The meaning of this signal depends on the mode of operation selected for Port A as follows:

(1) Output mode: The positive edge of this strobe is issued by the peripheral to acknowledge the receipt of data from the PIO.

(2) Input mode: The strobe is issued by the peripheral to load data from the peripheral into Port A of the PIO. Data is loaded into the PIO when this signal is active.

(3) Bidirectional mode: When this signal is active, data from the Port A output register is gated onto the Port A bidirectional bus. The positive edge of the strobe acknowledges the receipt of the data.

(4) Control mode: The strobe is inhibited internally.

ARDY — Register A ready (output, active high). The meaning of this signal depends on the mode of operation selected for Port A as follows:

(1) Output mode: This signal goes active to indicate that the Port A output register has been loaded and the peripheral data bus is stable and ready for transfer to the peripheral device.

(2) Input mode: This signal is active when the Port A input register is empty and is ready to accept date from the peripheral device.

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(3) Bidirectional mode: This signal is active when data is available in the Port A output register for transfer to the peripheral device. In this mode data is not placed on the Port A bus unless $\overline{ASTB}$ is active.

B0-B7 — Port B bus (bidirectional, tristate). This 8 bit bus is used to transfer data and/or status or control information between Port B of the PIO and a peripheral device. The Port B bus is capable of supplying 1.5 mA at 1.5 V to drive Darlington transistors. B0 is the least significant bit of the bus.

$\overline{BSTB}$ — Port B strobe pulse (input, active low). The meaning of this signal is similar to that of $\overline{ASTB}$ with the following exception:

In the Port A bidirectional mode, this signal strobes data from the peripheral device onto the Port A input register.

BRDY — Register B ready (output, active high). The meaning of this signal is similar to that of A Ready with the following exception:

In the Port A bidirectional mode this signal is high when the Port A input register is empty and ready to accept data from the peripheral device.

D0-D7 — CPU data bus (bidirectional, tristate). This bus is used to transfer all data and commands between the CPU and PIO. D0 is the least significant bit of the bus.

B/$\overline{A}$ SEL — Port B or A select (input, active high). This pin defines which port will be accessed during a data transfer between the CPU and PIO. A low level on this pin selects port A while a high level selects port B.

C/$\overline{D}$ SEL — Control or data select (input, active high). This pin defines the type of data transfer to be performed between the CPU and PIO. A high level on this pin during a CPU write to the PIO causes the data bus to be interpreted as a command for the port selected by the B/$\overline{A}$ SEL line. A low level on this pin means that the data bus is being used to transfer data between the CPU and the PIO.

CE — Chip enable (input, active low). A low level on this pin enables the PIO to accept command or data inputs from the CPU during a write cycle, or to transmit data to the CPU during a read cycle.

C — System Clock (input). The PIO uses the standard system clock to synchronize certain signals internally. This is a single phase clock.

$\overline{M1}$ — Machine Cycle 1 (input, active low). This signal from the CPU is used as a sync pulse to control internal PIO operations. When $\overline{M1}$ is active and the $\overline{RD}$ signal is active, the CPU is fetching an instruction from memory. Conversely, when $\overline{M1}$ is active and $\overline{IORQ}$ is active, the CPU is acknowledging an interrupt. In addition, the $\overline{M1}$ signal has two other functions within the PIO:

(1) $\overline{M1}$ synchronizes the PIO interrupt logic.

(2) When $\overline{M1}$ occurs without an active $\overline{RD}$ or $\overline{IORQ}$ signal the PIO enters a reset state.

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$\overline{IORQ}$	Input/Output request, (input, active low). The $\overline{IORQ}$ signal is used in conjunction with the B/$\overline{A}$ select, C/$\overline{D}$ select, $\overline{CE}$, and $\overline{RD}$ signals to transfer commands and data between the CPU and PIO.
$\overline{RD}$	Read cycle status (input, active low). The $\overline{RD}$ signal is used with B/$\overline{A}$ select, C/$\overline{D}$ select, $\overline{CE}$, and $\overline{IORQ}$ to transfer data between the CPU and PIO.

6.6 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finsih C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.7 Handling. MOS integrated circuits should be handled with certain precautions to avoid damage due to the accumulation of static charge. These NMOS devices are fabricated with a silicon gate technology, including input protection, which reduces the susceptibility to damage; however, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surfaces.

b. Ground test equipment and tools.

c. Handling of devices by the leads should be avoided.

d. Devices should be stored in conductive carriers.

e. Avoid uses of plastic, rubber, or silk in MOS areas.

6.8 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.3) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.9 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification shall functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Device type	Generic-industry type
01	Z80A PIO
02	Z80 PIO

6.10 Ordering guidance. Since the qualification requirements have been removed from the specification, orders may be placed immediately.

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/484A(USAF)	MICROCIRCUITS, DIGITAL, N-CHANNEL, SILICON GATE, MONOLITHIC, PARALLEL INPUT/OUTPUT CONTROLLER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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